

PADILLA BAY RESERVE, WA, USA

WMO: 998007

Lat: 48.467N

Lon: 122.467W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	26.2	14.0	11.2	26.7	18.9	14.3	29.5	26.9	47.0	24.5	47.0	4.4	60	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.1	74.8	64.8	72.5	63.5	70.5	62.4	66.2	73.1	64.5	70.9	63.3	69.3	8.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	86.7	69.9	61.8	82.5	68.2	60.3	78.3	66.7	30.7	73.0	29.6	71.2	28.6	69.4	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.5	18.8	16.5	17.6	80.4	4.0	2.7	14.7	82.4	12.4	84.0	10.1	85.5	7.2	87.5		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	39.6	40.2	44.0	47.4	52.9	57.5	60.6	60.7	57.2	50.2	43.5	40.0
	DBStd	9.18	5.86	6.32	4.91	3.83	3.79	3.67	3.24	3.16	4.05	5.09	6.39	6.14
	HDD50	1487	325	276	194	95	14	0	0	0	1	63	206	313
	HDD65	5655	789	694	651	528	376	227	141	137	234	460	644	774
	CDD50	1321	1	2	8	17	103	225	330	332	219	68	12	4
	CDD65	13	0	0	0	0	0	2	6	4	1	0	0	0
	CDH74	101	0	0	0	0	6	15	50	27	3	0	0	0
	CDH80	4	0	0	0	0	1	0	2	1	0	0	0	0
Wind	WSAvg	6.5	7.2	7.0	8.1	7.2	6.2	5.7	5.1	5.2	5.7	6.4	7.2	7.2
Precipitation	PrecAvg	43.5	5.3	3.4	4.1	3.5	3.1	2.5	1.1	1.4	2.2	4.4	6.7	5.6
	PrecMax	56.2	9.0	6.2	6.3	5.2	5.7	4.6	3.1	4.6	4.8	8.4	12.9	8.3
	PrecMin	33.7	2.9	0.5	1.5	1.0	1.1	0.5	0.1	0.1	0.7	1.3	2.5	2.9
	PrecStd	5.9	1.5	1.4	1.2	0.9	1.2	1.2	0.7	1.2	1.3	2.0	2.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	56.8	60.2	63.8	70.3	76.0	78.2	77.0	73.3	66.8	59.6	57.6
		MCWB	50.6	52.0	53.1	55.8	59.8	65.1	67.9	65.9	62.4	60.3	55.7	54.2
	2%	DB	51.8	52.9	56.6	59.8	66.5	72.5	75.1	74.1	70.1	63.0	56.6	53.2
		MCWB	48.8	49.2	50.6	52.8	58.8	63.0	65.3	64.4	62.3	58.1	53.3	50.6
	5%	DB	50.0	50.7	54.1	57.1	64.1	69.1	72.6	71.8	67.7	60.3	54.6	50.5
		MCWB	47.2	47.5	49.6	50.7	57.3	61.0	63.7	63.5	61.1	56.4	51.8	47.6
	10%	DB	47.9	48.8	52.1	54.8	61.5	66.5	70.4	69.8	65.4	58.5	52.6	48.5
		MCWB	45.6	46.0	48.2	49.3	55.6	59.1	62.6	62.4	60.0	55.2	50.1	46.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	53.4	54.5	56.9	61.9	67.3	69.6	67.6	65.9	61.9	56.6	54.9
		MCDB	52.9	56.5	58.2	62.8	67.3	75.1	76.9	74.9	70.3	65.7	58.4	57.4
	2%	WB	49.6	50.0	51.5	54.0	59.7	64.2	66.3	65.8	63.5	59.5	54.0	51.1
		MCDB	51.5	52.2	55.2	58.3	65.8	70.8	73.5	72.4	67.9	62.0	56.1	53.3
	5%	WB	47.8	48.2	50.2	51.8	57.9	61.6	64.4	64.3	62.1	57.4	52.2	48.4
		MCDB	49.7	50.2	53.7	55.6	63.3	68.1	71.6	70.5	66.2	59.6	54.2	50.2
	10%	WB	46.1	46.6	48.5	50.2	56.0	59.8	63.0	63.2	60.8	55.7	50.4	46.6
		MCDB	47.8	48.6	51.7	53.9	60.6	65.7	69.8	69.0	64.6	58.2	52.3	48.5
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.4	12.2	13.7	15.1	16.1	18.7	18.1	15.2	12.9	10.9	8.4
		MCDBR	9.2	11.7	14.3	17.7	19.9	20.8	22.4	21.1	18.8	14.1	10.7	8.4
		MCWBR	7.1	8.6	9.7	11.3	12.8	11.9	13.5	12.2	11.1	9.2	8.1	6.5
	5% WB	MCDBR	9.0	9.8	13.4	15.8	18.8	19.9	21.8	20.2	16.9	10.6	9.6	8.3
MCWBR		7.4	7.5	9.4	10.7	12.5	12.1	13.7	12.5	10.3	8.1	7.7	6.8	
Clear-Sky Solar Irradiance	taub	0.315	0.312	0.327	0.350	0.373	0.355	0.347	0.352	0.338	0.330	0.319	0.312	
	taud	2.535	2.561	2.536	2.457	2.411	2.475	2.524	2.518	2.552	2.566	2.542	2.499	
	Ebn at Noon	242	272	284	286	282	287	288	282	277	261	238	226	
	Edh at Noon	19	23	27	33	36	34	31	30	27	23	19	17	
All-Sky Solar Radiation	RadAvg	296	576	899	1361	1739	1902	2081	1755	1250	688	353	237	
	RadStd	39	76	105	127	137	180	146	114	102	76	54	31	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page